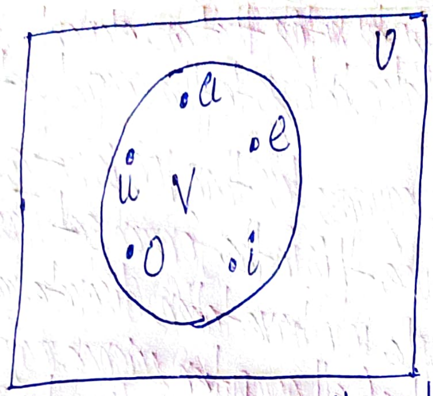

Figure 1: Venn diagram for the set of vowels

circle we indicate the elements of V with points. □

2.1.3 Subsets

The set A is a subset of B, and B is a superset of A, if and only if every element of A is also an element of B. We use the notation $A \subseteq B$ to indicate that A is a subset of the set B. If, instead, we want to stress that B is a superset of A, we use the equivalent notation $B \supseteq A$.

↳ Definition 3

- We see that $A \subseteq B$ if and only if the quantification $\forall x(x \in A \rightarrow x \in B)$ is true.
- Note that to show that A is not a subset of B we need only find one element $x \in A$ with $x \notin B$.

Important Rules

Showing that A is a Subset of B: If x belongs to A, then x also belongs to B.

Showing that A is not a Subset of B: Find a single $x \in A$ such that $x \notin B$.

Example 8: The set of all odd positive integers less than 10 is a subset of the set of all positive integers less than 10, the set of rational numbers is a subset of the set of real numbers, the set of all computer science majors at your school is a subset of the set of all students at your school, and the set of all people in China is a subset of the set of all people in China (that is, it is a subset of itself).